Computing Publications with Major Personal Contributions

Oliver Gutsche
May 10, 2026

1. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision
Aarrestad, Thea Klaeboe et al.
arXiv:2602.17582 | 2026-02-19

2. Evaluating Application Characteristics for GPU Portability Layer Selection
Atif, Mohammad et al.
arXiv:2601.17526 | 2026-01-24

3. Application of performance portability solutions for GPUs and many-core CPUs to track reconstruction kernels
Kwok, Ka Hei Martin et al.
EPJ Web Conf. 295 (2024) 11003 | arXiv:2401.14221 | DOI:10.1051/epjconf/202429511003 | 2024-01-25

4. The U.S. CMS HL-LHC R D Strategic Plan
Gutsche, Oliver et al.
EPJ Web Conf. 295 (2024) 04050 | arXiv:2312.00772 | DOI:10.1051/epjconf/202429504050 | 2023-12-01

5. A Ceph S3 Object Data Store for HEP
Smith, Nick et al.
EPJ Web Conf. 295 (2024) 01003 | arXiv:2311.16321 | DOI:10.1051/epjconf/202429501003 | 2023-11-27

6. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics
Atif, Mohammad et al.
arXiv:2306.15869 | 2023-06-27

7. Detector R D needs for the next generation e^+e^- collider
Apresyan, A. et al.
arXiv:2306.13567 | 2023-06-23

8. Automated Network Services for Exascale Data Movement
Balcas, Justas et al.
EPJ Web Conf. 295 (2024) 01009 | DOI:10.1051/epjconf/202429501009 | 2024

9. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics
Bockelman, Brian et al.
arXiv:2302.01317 | 2023-02-02

10. The Future of High Energy Physics Software and Computing
Elvira, V. Daniel et al.
arXiv:2210.05822 | DOI:10.2172/1898754 | 2022-10-11

11. Snowmass Computational Frontier: Topical Group Report on Experimental Algorithm Parallelization
Cerati, G. et al.
arXiv:2209.07356 | 2022-09-15

12. Portability: A Necessary Approach for Future Scientific Software
Bhattacharya, Meghna et al.

arXiv:2203.09945 | 2022-03-15

13. Coffea: Columnar Object Framework For Effective Analysis
Smith, Nicholas et al.
EPJ Web Conf. 245 (2020) 06012 | arXiv:2008.12712 | DOI:10.1051/epjconf/202024506012 | 2020-08-28

14. Using Big Data Technologies for HEP Analysis
Cremonesi, Matteo et al.
EPJ Web Conf. 214 (2019) 06030 | arXiv:1901.07143 | DOI:10.1051/epjconf/201921406030 | 2019-01-21

15. HEP Software Foundation Community White Paper Working Group – Data Organization, Management and Access (DOMA)
Berzano, Dario et al.
arXiv:1812.00761 | 2018-11-30

16. HEP Software Foundation Community White Paper Working Group - Data Analysis and Interpretation
Bauerdick, Lothar et al.
arXiv:1804.03983 | 2018-04-09

17. A Roadmap for HEP Software and Computing R D for the 2020s
Albrecht, Johannes et al.
Comput.Softw.Big Sci. 3 (2019) 7 | arXiv:1712.06982 | DOI:10.1007/s41781-018-0018-8 | 2017-12-18